Does Organization Matter? and Bargaining

Notes for Aulas 1–2 — Economics of Organizations and Incentives (MPE Insper)

Built on Milgrom & Roberts (1992), with Cabral (2017), Varian (2014, with Calculus) and Mas-Colell, Whinston & Green (1995)

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Abstract

Part I reads the Ford, GM and Toyota cases as answers to the two problems every organization faces, coordination and motivation, and then gives the mathematics of fit: complementarities, coherent systems, and the valley between them. Part II asks what bargaining decides. Under no wealth effects and complete information, the value maximization principle and the Coase theorem make the value-creating choice determinate, and bargaining power only divides the gains (Nash, Rubinstein). Private information breaks this: efficient trade can become infeasible, and incentive-efficient arrangements give up the small gains. Page numbers are Milgrom & Roberts' printed pages unless another book is named. Every worked example is original; book exercises are cited by number and page only.

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Part I

Aula 1 — Does Organization Matter? Ford × GM × Toyota

Syllabus item. *Aplicação: As organizações importam? Ford x GM x Toyota* (Topic 1). **Read.** Milgrom & Roberts (1992), Ch. 1 "Does Organization Matter?" (pp. 2–18); Ch. 4 "Coordination and Business Strategy" (pp. 106–113); Cabral (2017) §3.5 "Why are firms different?" (pp. 56–59).

Page numbers are M&R's printed pages unless another book is named. Facts that do not come from M&R carry their source inline; the stylized numbers are illustrative and checked by `aula-01-companions/check_companions.py` and the video's `check_numbers.py`. Three interactive companions in `Aula/aula-01-companions/` illustrate the formal part: `two_systems.html`, `valley.html` and `multiplier.html`.

1 The question

Ford had the technology, the scale and the cost advantage. GM beat it. Toyota was small, short of capital and technologically behind. It beat GM. M&R's answer, which this class builds up to:

> "organization and business strategy can be as important as technology, cost, and demand in determining a firm's success" (p. 16) — and what matters most is "the match of their strategies and their structures" (p. 16).

The class has two halves. First the cases, read as answers to two problems every organization faces, **coordination** and **motivation** (p. 17). Then the mathematics of **fit**: complementarities, why they make good systems coherent, and why coherent systems are hard to change one piece at a time.

2 Ford: one product, one system

M&R's picture of Ford is short and exact: "producing a single product (the Model T Ford) in high-volume factories using specialized equipment and rigidly disciplined labor paid according to the job being done" (p. 109). It was "an entirely different but equally coherent strategy that was appropriate for its time" and it took Ford past 50% of the US car market (p. 109). By 1921 the Model T held **55%** of the market, against **4%** for Chevrolet and **11%** for all GM brands together (p. 2).

The pieces, and why each raised the value of the others:

Choice	What it did	What it needed from the rest
One model, one colour	long, identical runs	a buyer who wanted a cheap standard car
Dedicated machines, moving line	labour per car collapsed	huge, steady volume
Narrow, repetitive tasks	unskilled labour could be used	close supervision, a pay premium to keep workers
High wage (the $5 day, 1914)	turnover fell	the productivity of the line to pay for it
Making its own parts	scale in components	the single design

History outside M&R (sources in `Videos/aula-01-does-organization-matter/provenance.md`): the Model T sold for about $850 in 1908 and about $260 by the mid-1920s; the moving chassis line (1913) cut assembly time from about 12.5 hours to 93 minutes by early 1914; 1913 labour turnover was 370% before the $5 day (Raff & Summers 1987; the turnover figures are marked *to confirm* there). The wage story is efficiency wages, M&R Ch. 8 (p. 250), outside this course's core.

The weakness was the same as the strength. Ford's 1920 answer to the recession was a price cut of about 25%, which GM, with higher costs, could not match (p. 2). Cost leadership worked while buyers wanted the same car. It stopped working when they wanted different cars.

3 GM: variety, and the organization it demanded

Sloan took over a GM in crisis in 1921: unsold inventories piling up, sales down 75% between summer and fall 1920, divisions "competed mostly with one another" (pp. 2–3).

Strategy. A car "for different segments of the market": Cadillac at the top, each division a lower-income segment, Chevrolet below the Model T's price (p. 3).

Why the strategy required a new organization. Variety multiplied the decisions: new designs, dealers, market information per segment, separate factories and supplies. At the same time the divisions had to be coordinated: not compete too much, share ideas, coordinate R&D, standardize parts "enough to achieve economies of scale in parts production" (p. 3). Neither Ford's centralized one-product organization nor the old GM, "a collection of car companies and suppliers operating without any central direction", could do this (p. 3). The old accounting system had not even charged divisions for the inventories they built up, so they kept producing into the recession (p. 3).

The multidivisional form (p. 3):

- divisions keep operating autonomy, each with its own management team and market segment;
- a strong central office does *not* run day-to-day operations; it audits and evaluates divisions, plans and coordinates strategy, and keeps research, legal and financial functions;
- the centre checks internal accounting prices against market prices, so supply divisions can be judged "as if each were a separate firm" (p. 4).

Henry Ford mocked the "great big chart" where a message takes six weeks to travel (p. 4). The result: from 1927 to 1937 Ford lost $200 million and GM earned over $2 billion; by 1940 GM had 45% of the market, Ford 16% (p. 4).

GM's system fit together too. Shared components across models let GM afford specialized equipment for chassis, engines and brakes (p. 5); its volumes fully used efficient parts plants, so it integrated vertically (p. 5), and it made suppliers compete on price (p. 6). The price of specialization: major redesigns only about every 12 years in the 1950s and 1960s (p. 5).

4 Toyota: small, poor, and coherent in the other direction

Early-1950s Toyota had a "drastic lack of capital" and a scale too small to match US costs (p. 4). It first tried to copy mass production, then built a different system under Eiji Toyoda and Taiichi Ohno (p. 4).

Practice (M&R pp. 4–6)	Why it followed from Toyota's position	What it then required
Just-in-time / kanban: no buffer inventories	inventory systems have large scale economies Toyota could not reach	every step reliable; defects caught at once
Operators maintain and repair their own machines	without buffers, a breakdown stops everything	broader skills, authority on the floor
Flexible equipment, quick resets, robots by the 1960s	too small to dedicate machines to one model	frequent model changes became cheap: redesign twice as often as GM
Outside suppliers for complex components	could neither reach efficient scale in parts nor afford to own parts makers	close day-to-day coordination with suppliers
Few suppliers, long-term relations	high-level components that must fit, frequent redesigns	suppliers willing to invest in Toyota-specific skills and machines

M&R's summary of the logic: "If low inventories were to be achieved, the rest of the organization had to emphasize reliability, quick response to machine breakdowns, quick and close communications with suppliers, stable production plans, and other features that substitute for the buffer function of inventories" (p. 17). Cabral adds why the system was hard to copy: "the system involved so many different aspects that it was difficult to pin down what the crucial success factors were" (Cabral 2017, §3.5, p. 57).

4.1 Ford, GM, Toyota side by side

Dimension	Ford (1910s)	GM (1920s on)	Toyota (1950s on)
Product line	one model	a ladder of segments	several models, frequent redesign
Equipment	dedicated	specialized, shared components	flexible
Inventories	buffers	buffers	none (just-in-time)
Suppliers	made its own parts	vertically integrated; price competition	few, long-term, outside
Decisions	centralized in Henry Ford	divisions operate; centre audits and sets strategy	problem solving on the floor
Workers	narrow tasks, high wage	(not developed in M&R)	multi-skilled, maintain machines

Each column is coherent. None is a list of independent "best practices".

5 Three more cases, one lesson each

M&R widen the claim beyond cars (pp. 6–16). Each case isolates one piece of the argument.

- **Hudson's Bay vs. North West Company** (pp. 6–9): **incentives and decision rights.** HBC had half the import costs, a royal monopoly and posts on the bay. It was run from London, with 15-month communication lags, flat salaries and rules enforced by flogging. NWC traders paddled an extra 1,500 miles. Its "wintering partners" decided on the spot and shared the profits. By about 1810 NWC had nearly 80% of the trade (p. 8). HBC recovered

only by copying the *organization*: inland posts, half of profits to field officers, more freedom of action (pp. 8–9).

- **Salomon Brothers** (pp. 9–12): **incentives that are too individual.** Near piece-rate bonuses made traders work hard and take risks but discouraged sharing information across departments (p. 11). The 1990 reform paid part of each bonus in stock held in trust for five years; the announcement raised the firm's value by 7–10% (p. 12). FFT 1.3 (p. 18) asks what the 1991 scandal says about that scheme.
- **Eastern Europe** (pp. 12–16): **information, incentives and interdependent reforms.** Central planning replaced prices with vertical information flows (p. 13). The **ratchet effect**: a manager who beats the quota gets a higher quota, so nobody beats it; capacity is hidden and resources hoarded (p. 14). Reform pieces depend on each other (property rights, capital markets, contract enforcement), so "the worst outcome might be some halfway compromise" (p. 16): the country that moved only its trucks to the other side of the road. The ratchet effect returns in Aula 16 (M&R p. 232).

6 The patterns: coordination, motivation, and fit (pp. 16–17)

M&R draw four patterns out of the cases.

1. **Organization and strategy can matter as much as technology, cost and demand.** Ford lost to GM despite better technology and scale; GM lost share to a smaller, weaker Toyota; HBC lost to NWC (p. 16).
2. **Incentives matter.** HBC employees got no share of extra profits; GM's divisions were not charged for inventories; Salomon rewarded individuals and so discouraged cooperation; planned economies "ran afoul of incentives at every turn" (p. 16).
3. **Put authority where the information is.** Salomon's traders, GM's divisions, Toyota's operators, NWC's wintering partners (p. 17).
4. **Incentives and delegated authority are complements.** Delegation is worth little "unless the decision makers share the organization's objectives", and incentives matter most "when more initiative is expected from employees". In M&R's words: "incentives and delegated authority are complements: each makes the other more valuable" (p. 17).

And the organizing pair for the course:

> *"The study of organization is not about how berries are arranged on a tree of authority but about how people are coordinated and motivated to get things done" (p. 16).*

	Coordination: getting the plans to fit (information)	**Motivation**: getting people to carry them out (interests)
Ford	one product, one centre, the line sets the pace	high wage, close supervision
GM	divisions per segment; centre plans strategy, shared parts	divisions judged on profit, internal prices checked against the market
Toyota	just-in-time signals, close supplier links	multi-skilled teams, long-term supplier relations
HBC → NWC	decisions moved to the field	profit shares for those deciding

Course map: coordination in Aulas 6–8 (M&R Ch. 3–4), motivation in Aulas 9–16 (Ch. 5–7).

7 Complementarities: the mathematics of fit (M&R Ch. 4, pp. 108–109)

7.1 Definition

The market definition: two inputs are complements if a fall in the price of one raises demand for the other. M&R use a "more inclusive" definition that also covers internal activities:

Definition: Complementary activities (M&R p. 108)

"Several activities are mutually complementary if doing more of any one activity increases (or at least does not decrease) the marginal profitability of each other activity in the group" (p. 108).

Smooth version (footnote 8, p. 108). With activity levels $x = (x_1, \ldots, x_n)$ and profit $\pi(x)$:

$$\frac{\partial^2 \pi}{\partial x_i \, \partial x_j} \geq 0 \qquad \text{for all } i \neq j.$$

Discrete version: increasing differences. For two yes/no choices $x, y \in \{0, 1\}$:

$$\underbrace{\pi(1,1) - \pi(0,1)}_{\text{gain from } x \text{ when } y=1} \geq \underbrace{\pi(1,0) - \pi(0,0)}_{\text{gain from } x \text{ when } y=0}.$$

Result: increasing differences is symmetric and equals supermodularity

Rearranging the same inequality three ways:

$$\begin{aligned} \pi(1,1) - \pi(0,1) \geq \pi(1,0) - \pi(0,0) &\iff \pi(1,1) + \pi(0,0) \geq \pi(1,0) + \pi(0,1) \\ &\iff \pi(1,1) - \pi(1,0) \geq \pi(0,1) - \pi(0,0). \end{aligned}$$

The last line says the gain from y is larger when $x = 1$ (symmetry). The middle line is **supermodularity**: the two "matched" corners together beat the two "mixed" ones.

Example: The video's 2×2 table

$\pi(0,0) = 0$, $\pi(1,0) = -2$, $\pi(0,1) = -1$, $\pi(1,1) = 5$.

- Gain from x when $y = 0$: $-2 - 0 = -2$; when $y = 1$: $5 - (-1) = 6$. Increasing differences hold.
- Each practice **alone loses money**, and the pair gains 5. The cross effect is $6 - (-2) = 8$.
- A firm that tests each practice separately rejects both.

*Complementarity says nothing about whether each practice is good on its own. It says each is **better where the other is present**.*

7.2 Why complements move together

One choice, one parameter (Topkis, stated). If $x^*(t)$ maximizes $f(x, t)$ and f is concave in x, differentiate the first-order condition $f_x(x^*(t), t) = 0$ with respect to t:

$$f_{xx}\, x^{*\prime}(t) + f_{xt} = 0 \implies x^{*\prime}(t) = -\frac{f_{xt}}{f_{xx}} \geq 0 \quad \text{whenever } f_{xt} \geq 0,$$

since concavity gives $f_{xx} < 0$ at an interior optimum where the implicit function theorem applies.

Theorem (Topkis)

If f has increasing differences in (x, t), the optimal x is non-decreasing in t. No derivatives or concavity are needed.

Milgrom & Roberts (1990, *AER* 80(3), 511–528) used it to explain why the elements of modern manufacturing were adopted together. Stated, not proved: the lattice proof is beyond the course.

Two choices, the systems effect. Let x be delegated authority and y incentive intensity (M&R's p. 17 pair), with

$$\pi(x, y) = tx + sy + b\,xy - \tfrac{c}{2}x^2 - \tfrac{c}{2}y^2, \qquad \frac{\partial^2 \pi}{\partial x\,\partial y} = b.$$

First-order conditions:

$$\pi_x = t + by - cx = 0, \qquad \pi_y = s + bx - cy = 0 \quad \Longleftrightarrow \quad t + by = cx, \qquad s + bx = cy.$$

From the first, $x = (t+by)/c$. Substituting into the second, $s+b(t+by)/c = cy$, i.e. $cs+bt+b^2y = c^2y$, so $y^* = (cs + bt)/(c^2 - b^2)$; by symmetry of the two conditions,

$$x^* = \frac{ct + bs}{c^2 - b^2}, \qquad y^* = \frac{cs + bt}{c^2 - b^2}, \qquad \text{a maximum iff } c > |b|.$$

The Hessian is $\left(\begin{smallmatrix} -c & b \\ b & -c \end{smallmatrix}\right)$: negative definite iff $-c < 0$ and $c^2 - b^2 > 0$, which together are $c > |b|$.

How delegation responds to more valuable local information t:

$$\frac{\partial x^*}{\partial t} = \frac{c}{c^2 - b^2} = \frac{1}{c} \cdot \frac{1}{1 - (b/c)^2} = \frac{1}{c}\Big(1 + \big(\tfrac{b}{c}\big)^2 + \big(\tfrac{b}{c}\big)^4 + \cdots\Big), \qquad \frac{\partial y^*}{\partial t} = \frac{b}{c^2 - b^2}.$$

- $1/c$ is the response with y held fixed. The geometric series is M&R's footnote 8 in numbers: more x raises the return to y, more y raises the return to x, "possibly leading to another round of increases" (p. 108).
- t never enters the condition for y, yet y^* moves: with $b > 0$ incentives rise when delegation becomes more valuable.
- At $c = 2$, $b = 1$: multiplier $1/(1 - 1/4) = 4/3$. With t from 1 to 3 ($s = 1$), x^* goes from 1 to $7/3$ instead of from 0.5 to 1.5.
- With **substitutes** ($b < 0$) the multiplier is still above one, because $(b/c)^2 > 0$: incentives fall, which raises delegation's return again. Only the sign of $\partial y^*/\partial t$ flips.

The companion `multiplier.html` *(The Complementarity Multiplier) illustrates this: moving t shows the staircase of rounds, $b = 0$ makes the multiplier fall to 1, and pushing $|b|$ toward c makes it explode until the optimum vanishes.*

7.3 Substitutes are complements upside down

If $\partial^2 f/\partial x\,\partial y \le 0$ (buffer inventories and floor-level problem solving are substitutes: each makes the other less needed), define $z = -y$ and $g(x, z) = f(x, -z)$. Then $g_{xz} = -f_{xy} \ge 0$. Every result above applies to (x, z). "Many buffers, narrow jobs" (Ford) and "no buffers, problem solvers" (Toyota) are both coherent; the mixtures are not.

7.4 Strong complementarity and design attributes (p. 109)

Complementarity makes the choices **predictable**: high levels go together. M&R add a second, separate feature, the **high cost of failing to fit**, and call a group *strongly complementary* when raising some activities "greatly increases the returns" to raising the others; then "design attributes are always present" (p. 109). The box "Modern Manufacturing Strategy" (pp. 109–110) traces one chain:

> small demand per product → no finished-goods inventories → close sales–factory communication → short production runs → frequent changeovers → flexible equipment → multi-skilled workers paid for skills, not piece rates → machine-time cost accounting.

8 More than one coherent system (M&R Ch. 4, pp. 111–113)

8.1 Figure 4.3, made exact

M&R's figure: a **marketing manager** chooses the number of product varieties n; a **manufacturing manager** chooses the batch size b (p. 111). More varieties at a fixed batch size raise inventories, so the larger the batch, the fewer varieties are worth offering; more varieties raise the marginal inventory cost of a larger batch, so the more varieties, the smaller the best batch (p. 111). Both best replies slope down, and they can cross more than once (p. 112).

M&R draw the curves without a formula. A form that produces them, with equipment $\theta \in \{\text{specialized}, \text{flexible}\}$ added (pp. 5, 113):

$$\pi(n, b, \theta) = mQ + A \ln n - \frac{S_\theta Q}{b} - \frac{h\, n\, b}{2} - \Delta c_\theta\, Q, \qquad n \geq 1,$$

where A is the taste for variety, Q total volume, $S_\theta Q/b$ the changeover cost (Q/b changeovers), $hnb/2$ the cost of holding average inventory $b/2$ for each of n models, and Δc_θ the unit-cost premium of flexible equipment (zero for specialized).

- **Manufacturing's reply (the EOQ formula).** $\pi_b = S_\theta Q/b^2 - hn/2 = 0$ gives $b^*(n) = \sqrt{2S_\theta Q/(hn)}$, decreasing in n.
- **Marketing's reply.** $\pi_n = A/n - hb/2 = 0$, so $A/n = hb/2 \Rightarrow n^*(b) = \max\{1,\ 2A/(hb)\}$, decreasing in b.
- **Cross effect.** $\partial^2\pi/\partial n\, \partial b = -h/2 < 0$: variety and batch size are substitutes, so variety and **small batches** are complements (§7.3).
- **Within one equipment type.** Substituting $n = 2A/(hb)$ into $SQ/b^2 = hn/2$ gives $SQ/b^2 = A/b$, so $b^* = SQ/A$ and $n^* = 2A^2/(hSQ)$. At the optimum $SQ/b^* = A$ and $hn^*b^*/2 = A$, so $\pi^* = mQ + A\ln n^* - 2A - \Delta c\, Q$. Corner $n^* = 1$ (Ford's single model) when $2A^2 \leq hSQ$.
- **The equipment switch.** At the EOQ batch the two batch-related costs are equal, so $SQ/b^* + hnb^*/2 = \sqrt{2SQhn}$. Flexible equipment is cheaper once $\sqrt{2Qhn}\,(\sqrt{S_s} - \sqrt{S_F}) \geq \Delta c\, Q$, i.e. once n exceeds
$$n_s = \frac{(\Delta c\, Q)^2}{2Qh\left(\sqrt{S_s} - \sqrt{S_F}\right)^2}.$$
Manufacturing's reply **jumps down** there.

With the companion's calibration ($A = 20$, $Q = 100$, $h = 1$, $S_s = 50$, $S_F = 2$, $\Delta c = 1$, $m = 2$):

System	n^*	b^*	Profit
Specialized (Ford-type)	1 (corner)	100	100.0
Flexible (Toyota-type)	4	10	87.7

Here $\sqrt{S_s} - \sqrt{S_F} = 5\sqrt{2} - \sqrt{2} = 4\sqrt{2}$, so $n_s = 100^2/(2 \cdot 100 \cdot 32) = 1.5625$. Manufacturing's reply is $b = 100/\sqrt{n}$ for $n \le n_s$ and $b = 20/\sqrt{n}$ beyond it, dropping from 80 to 16 at n_s. Marketing's reply is $n = 1$ for $b \ge 40$ and $n = 40/b$ below.

Both are coherent: at each, neither manager wants to move. Between them, profit along manufacturing's reply falls to **83.9**. Raise A to 30 and the flexible system earns 105.9, better than 100, yet the valley is still there (88.4).

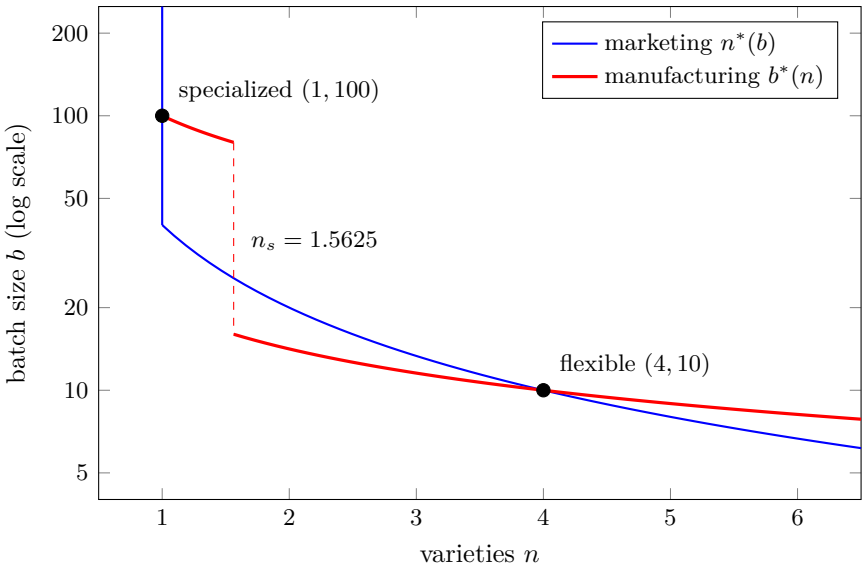


Figure 1: Figure 4.3 made exact, with $A = 20$, $Q = 100$, $h = 1$, $S_s = 50$, $S_F = 2$, $\Delta c = 1$. Marketing's reply is $n^*(b) = \max\{1, 2A/(hb)\}$ (vertical at $n = 1$ for $b \ge 40$); manufacturing's reply $b^*(n) = \sqrt{2SQ/(hn)}$ jumps from 80 to 16 at $n_s = 1.5625$, where it switches from specialized to flexible equipment. The two crossings are the two coherent systems.

8.2 What the figure teaches

- **Alignment is not optimality.** Each crossing is "a coherent strategy" (p. 112); "only one combination is actually profit maximizing" (p. 112). Ford's crossing was best for the early century; the many-varieties, small-batch crossing is best for many industries now (p. 112).
- **Local adjustment cannot find the global shift.** Managers can track the *local* movement of their own crossing, "but recognizing that a radical strategic shift is desirable is likely to require information that neither manager has" (p. 113). Smooth changes in the curves can hide a jump in which crossing is best (p. 112).
- **So the choice among coherent patterns is a job for central management** (p. 113, p. 115), with innovation attributes, because the information is outside the organization. This is Aula 7's argument, met here first.

A careful reading of footnote 9

M&R picture the profit surface as "a mountain with three peaks, one at each of the intersections" (p. 112). With two *smooth* decreasing best replies that cross three times, the middle crossing is a **saddle**, not a peak: there the replies cross the "wrong" way, $\pi_{nb}^2 > \pi_{nn}\pi_{bb}$, and the Hessian is indefinite. The outer two crossings are the peaks. The book's point is unaffected: several coherent systems, one best.

The companion `two_systems.html` *(Two Coherent Systems) illustrates this: its plate 1 is Figure 4.3, its plate 2 the "ridge line" of footnote 9, and its presets* More taste for variety *and* Cheaper flexibility *move each system a little while the best one switches.*

9 The valley, and the plant that tested it

9.1 Three practices on a cube

Take three Toyota practices from §4: just-in-time J, multi-skilled teams T, long-term suppliers S, each 0 or 1. The class's stylized payoff:

$$\pi(J,T,S) = 10 - 4J - 2T - S + 4JT + 5JS + 2TS + 2JTS.$$

Corner	000	100	010	001	110	101	011	111
Profit	**10** (Ford)	6	8	9	8	10	9	**16** (Toyota)

- Supermodular: the pairwise gains $4 + 2S$, $5 + 2T$, $2 + 2J$ are all positive.
- **Every first step loses**: 6, 8, 9 $<$ 10. Ford's corner is a local peak.
- **Every adoption order dips.** Of the six orders from 000 to 111, the best one still falls to 9 before reaching 16.
- 111 is also a local peak (its neighbours earn 9, 10, 8). Two peaks, and marginal analysis cannot tell them apart: the package gain is +6; the best single step is −1.

Complementarity tells you ***where*** *the optimum is (all practices together). It does not make the road uphill. It* ***creates*** *the valley, because a practice is worth least where its complements are missing.*

The companion `valley.html` *(The Valley Between Systems) illustrates this: its preset* No interactions *shows "best practices" thinking (profits add up and the package loses), and* No stand-alone losses *lets some order climb without a dip.*

9.2 NUMMI: the same workers under two systems

History outside M&R (sources and status in `Videos/aula-01-does-organization-matter/provenance.md`):

- GM closed its Fremont, California plant in 1982. It reopened in 1984 as **NUMMI**, a GM–Toyota joint venture run on Toyota's system, and over 85% of its first workforce were the laid-off GM workers.

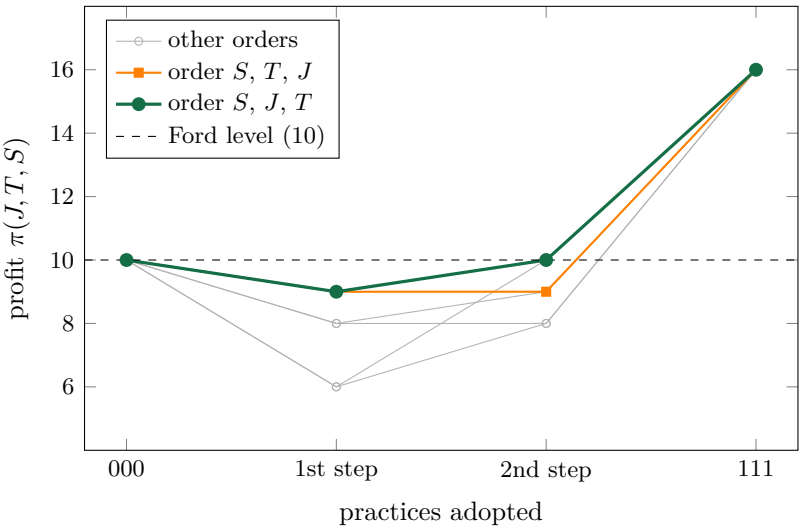


Figure 2: Profit along the six adoption orders from 000 (Ford) to 111 (Toyota), read off the corner table. Every order dips below the Ford level 10; the best orders, which adopt S first, fall to 9 before reaching 16.

- 1987 assembly hours per car: **31** at a comparable GM plant (Framingham), **19** at NUMMI, 16 at Toyota's Takaoka plant; assembly defects per 100 cars 135, 45 and 45. These figures come from a secondary reproduction of Womack, Jones & Roos (1990) and are marked *to confirm against the book*.
- GM owned half the plant and its managers could study it, yet its efforts to spread the system piece by piece across its US plants "proved largely unsuccessful". The exception was Brazil, where NUMMI-trained managers ran the transfer.

Read with §9.1: Toyota moved the whole system at once (the jump across the valley). GM, adopting practice by practice, kept landing on the low corners.

9.3 Evidence beyond one plant

A test regression: $y = \beta_0 + \beta_1 x + \beta_2 z + \beta_3 xz + \varepsilon$. The effect of x is β_1 when $z = 0$ and $\beta_1 + \beta_3$ when $z = 1$: β_3 is the cross effect. Ichniowski, Shaw & Prennushi (1997, *AER* 87(3), 291–313), on 36 steel finishing lines: the bundle of innovative HR practices raises productivity; single practices adopted in isolation do not.

Correlated adoption is not proof

Firms that adopt practices together may differ in other ways, so correlated adoption alone does not prove complementarity.

10 Common mistakes

Mistake	Correction
"Copy best practices one at a time"	Profit is not additive; with complements, a practice adopted alone can lose money (§7.1, §9.1).
"Complements means each is good"	Complements means each is *better with* the other. Both can lose alone (the 2×2 table).
"A coherent system is the best system"	Coherence is a local property. Several coherent systems coexist; only one is best (§8.2).
"If it is better, managers will drift to it"	Marginal adjustment stays at a local peak; reaching the other needs a coordinated jump (pp. 112–113).
Using the market definition of complements (cross-price) for internal choices	M&R's definition is about marginal profitability, $\partial^2\pi/\partial x_i \partial x_j \geq 0$ (p. 108).
"Substitutes have a multiplier below one"	In the two-choice model the own response is amplified for either sign of b; only the other choice's direction changes (§7.2).
Reading organization as the org chart	"not about how berries are arranged on a tree of authority but about how people are coordinated and motivated" (p. 16).

11 Exercises (number and page only)

Exercise	Where	Practises
M&R Food for Thought 1.1	p. 18	splitting a business's decisions between centre and units; coordination vs. motivation
M&R Food for Thought 1.2	p. 18	the same split in a non-business organization
M&R Food for Thought 1.3	p. 18	the Salomon incentive scheme (§5) against a later scandal
Cabral 1.1	p. 11	competition and performance
Cabral 3.14	p. 63	sustained competitive advantage (§3.5)
Cabral 3.20	p. 64	productivity dispersion across firms (applied, data)

For the formal part (§7–§9) there are no book exercises in M&R Ch. 1; use the companions' presets as problems: predict the read-out, then move the control.

Sources

- Milgrom, P. & Roberts, J. (1992). *Economics, Organization, and Management*. Prentice-Hall. Ch. 1 (pp. 2–18), Ch. 4 (pp. 106–115). Read from `Livros/` on 2026-10-09.
- Milgrom, P. & Roberts, J. (1990). "The Economics of Modern Manufacturing: Technology, Strategy, and Organization". *American Economic Review* 80(3), 511–528.
- Cabral, L. (2017). *Introduction to Industrial Organization*, 2nd ed. MIT Press. §3.5, pp. 56–59.
- Ichniowski, C., Shaw, K. & Prennushi, G. (1997). *American Economic Review* 87(3), 291–313.
- Ford and NUMMI history: `Videos/aula-01-does-organization-matter/provenance.md` (each claim with its source and status).

Part II

Aula 2 — Bargaining

Read. M&R Ch. 2, "Wealth Effects, Value Maximization and the Coase Theorem" (pp. 35–39); Ch. 5, "Achieving Commitment" (pp. 133, 139–140), "Private Information and Precontractual Opportunism" and "Measurement Costs and Investments in Bargaining Position" (pp. 140–149). Varian (2014, with Calculus) §30.7 "Bargaining" and "The Ultimatum Game" (pp. 581–583), §35.2 (p. 670). MWG Ch. 9, Appendix A (pp. 296–299). Cabral (2017) §5.1 (pp. 96–99).

In the draft schedule, Bargaining takes two classes: Aula 2 covers §12–§16 (value maximization, Coase, the division of the gains), and Aula 3 covers §17–§22 (alternating offers, commitment, private information, bargaining costs).

12 The question: is bargaining indeterminate?

Every organization runs on agreements: a supplier contract, a division of profits among partners, a transfer price between divisions, a wage. Each agreement is reached by some form of bargaining. M&R open the topic with the traditional view that "the outcomes of bargaining among small numbers of people are essentially indeterminate": bargaining strength, credibility, guile, patience and strategic insight combine with initial conditions to decide the outcome, and predicting it "was too much for economic analysis" (p. 140).

This Part gives M&R's answer in three steps.

1. **What is created is determinate.** When preferences show no wealth effects, every efficient agreement chooses the actions that maximize the parties' total value (the *value maximization principle*). Bargaining power decides only the money transfers (the *Coase theorem*). Sections 13–15.
2. **Who gets what can be modelled.** The Nash solution and the Rubinstein alternating-offers game turn "bargaining power" into threat points, patience and the right to propose. Under complete information these models predict immediate, efficient agreement. Sections 16–18.
3. **Private information and bargaining costs break efficiency.** When each side knows something the other does not, the full-information efficient outcome can be infeasible, and even the best feasible arrangement gives up some gains. Time, haggling and information spent on shifting the division are wasted. Sections 19–22.

Throughout, a decision has two parts: y, the *real* choice (what is produced, who owns an asset, whether a project is built), and $x = (x_1, \ldots, x_N)$, the money transfers among the parties.

13 Wealth effects and the value index

A *wealth effect* is a change in choices caused by a change in wealth: a poorer person may be unable to afford an option, or may trade off risk differently (p. 35).

Definition: No wealth effects (p. 35)

A decision maker shows *no wealth effects* with respect to a set of decisions Y when, for any two decisions $y_1, y_2 \in Y$:

(i) there is a definite amount of money Δx that would just compensate him or her for a

switch from y_1 to y_2;

(ii) that amount does not change if he or she is first given extra wealth;

(iii) he or she has enough money to absorb any wealth reduction needed to pay for a switch from the less preferred to the more preferred option.

Proposition 1 (the value index, M&R p. 36 and n. 7)

Under no wealth effects, preferences over (money x, decision y) can be represented by

$$u(x, y) = x + v(y),$$

where $v(y)$ is the cash-equivalent value of y. The compensating transfer for a switch from y_1 to y_2 is $\Delta x = v(y_1) - v(y_2)$, the same at every initial wealth M.

Why. With $u = x + v(y)$, the person is indifferent between (M, y_1) and $(M + \Delta x, y_2)$ exactly when $M + v(y_1) = M + \Delta x + v(y_2)$, so $\Delta x = v(y_1) - v(y_2)$: it exists (condition i) and does not depend on M (condition ii). Condition (iii) holds as long as M exceeds the largest difference $|v(y) - v(y')|$. Conversely, M&R's Q 2.4 (p. 53) asks you to show that conditions like (i)–(ii) force this form.

Where it fails. M&R contrast two cases (p. 35–36). A corporation that finds itself richer after a rise in its asset values would plausibly keep the same prices and hurdle rates: no wealth effects. A worker who wins the lottery buys a house and quits: large wealth effects. Condition (iii) fails when the transfer is large relative to the person's resources, as with a nuclear-plant worker made financially liable for his or her mistakes (p. 36). The rule of thumb: the assumption is least safe "when the decision makers are individuals and when large cash transfers or significant changes in personal living conditions are involved" (p. 36).

Example: With wealth effects, who holds the right changes what is done

A plant gains 50 (in money) from dumping waste in a river. A resident with wealth $w = 100$ has utility $u = \ln x + a\,y$, where $y = 1$ if the river is clean and $a = 0.5$. This is not of the form $x + v(y)$: the money value of a clean river depends on wealth.
Willingness to pay for a clean river (resident owns nothing, must buy it): $\ln(w - P) + a = \ln w \;\Rightarrow\; P = w(1 - e^{-a}) \approx 39.35$.
Willingness to accept for giving the river up (resident owns the right): $\ln(w + A) = \ln w + a \;\Rightarrow\; A = w(e^{a} - 1) \approx 64.87$.
If the plant holds the right, the resident can offer at most $39.35 < 50$: the plant dumps. If the resident holds it, the plant would have to pay at least $64.87 > 50$: the river stays clean. The *real* outcome y depends on the assignment of rights. With wealth effects, the gap between willingness to pay and to accept makes "the efficient choice" depend on who starts with what, and the clean separation of the next section disappears.

14 The value maximization principle

The Value Maximization Principle (M&R p. 36)

"An allocation among a group of people whose preferences display no wealth effects is efficient only if it maximizes the total value of the affected parties. Moreover, for any inefficient allocation, there exists another (total value maximizing) allocation that all of the parties

strictly prefer."

Setting. N people with $u_n(x_n, y) = x_n + v_n(y)$. Decision y generates net cash $P(y)$ (positive for an investment, negative for a public good that costs money), and the transfers must add up: $x_1 + \cdots + x_N = P(y)$. The *total value* of an allocation is

$$\sum_n \left[x_n + v_n(y)\right] \;=\; P(y) + \sum_n v_n(y) \;\equiv\; W(y),$$

which depends on y alone: changing the x's moves value between people one for one and never changes the total.

Proposition 2 (efficiency ⇔ value maximization)

An allocation $(y, x_1, \ldots, x_N)$ is Pareto efficient if and only if $y \in \arg\max_{y'} W(y')$.

Proof. ($\Leftarrow$) Let y maximize W and suppose another allocation (y', x') made everyone at least as well off and someone strictly better off. Summing utilities, $W(y') = \sum_n [x'_n + v_n(y')] > \sum_n [x_n + v_n(y)] = W(y)$, contradicting the maximality of $W(y)$.

($\Rightarrow$) Suppose y does not maximize W: some y^* has $G \equiv W(y^*) - W(y) > 0$. Give each person the transfer

$$x_n^* = x_n + v_n(y) - v_n(y^*) + \frac{G}{N}.$$

Then $u_n(x_n^*, y^*) = x_n + v_n(y) + G/N > u_n(x_n, y)$ for every n. The transfers add up: using $\sum_n x_n = P(y)$ and $G = P(y^*) + \sum_n v_n(y^*) - P(y) - \sum_n v_n(y)$,

$$\sum_n x_n^* = P(y) + \sum_n v_n(y) - \sum_n v_n(y^*) + G = P(y^*).$$

So every inefficient allocation is strictly Pareto dominated by a value-maximizing one, which is the second sentence of the principle. □

M&R's Q 2.3 (p. 53) asks for this proof; the hint there is the equal-split construction used above. The proof uses condition (iii): the transfers x_n^* must be affordable.

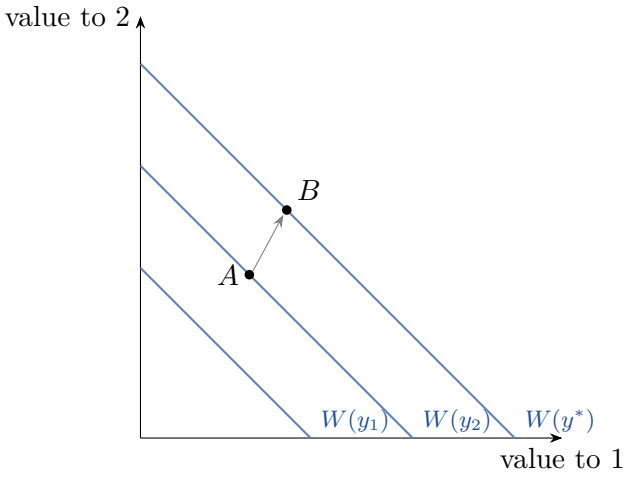


Figure 3: M&R's Figure 2.1 (p. 37). Each line is the set of value pairs reachable with one decision y as the transfers vary; lines are at 45° because transfers move value one for one. Any point A on a lower line is dominated by some B on the highest line; points on the highest line are undominated.

When the principle applies, "the issue of the distribution of value is completely separable from the issue of how value is created" (p. 38). Efficiency picks y; everything else (bargaining power, patience, initial ownership) only moves the x's.

The limit (M&R n. 10, p. 37). With wealth effects, an allocation that maximizes the sum of utilities is still efficient, but there can be efficient allocations that do not: value maximization becomes *sufficient* but not *necessary* for efficiency. The utility possibility frontier is then curved rather than a 45° line, and different points on it can use different y's (as in the river example above).

Range of application (p. 38). The variable y can be the money spent on parks or libraries, the identity of the person assigned overtime, the owner of water rights, the brand of computer adopted, or the site of a new office building. In each case the principle says the group should, and if it bargains efficiently will, pick the y with the largest total value, then settle shares by transfers.

15 The Coase theorem

The Coase Theorem (M&R p. 38)

"If the parties bargain to an efficient agreement (for themselves) and if their preferences display no wealth effects, then the value-creating activities (y) that they will agree upon do not depend on the bargaining power of the parties or on what assets each owned when the bargaining began. Rather, efficiency alone determines the activity choice. The other factors can affect only decisions about how the costs and benefits are to be shared (x)."

The theorem is the value maximization principle plus one behavioural assumption: the parties reach an agreement with no further mutual gains. Its organizational corollary is M&R's foundation for the transaction-costs approach: for a given production plan, "the efficient organization for that plan is the one that minimizes transaction costs" (p. 38), where transaction costs include writing contracts, supervising, enforcing and resolving disputes.

Example: The steel plant and the river (after Cabral, pp. 97–98)

Cabral's setting: a steel plant can dump waste in a river, harming residents downstream; society can give the right to clean water to the residents or the right to dump to the plant. Put numbers on it: dumping is worth 100 to the plant and costs the residents 60, everyone has no wealth effects, and bargaining is costless.

- *Plant holds the right.* The residents would pay at most 60 to stop dumping; the plant needs at least 100. No deal: the plant dumps, no money changes hands.
- *Residents hold the right.* The plant can pay anything in $[60, 100]$ for permission. With an even split of the 40 gain (the Nash solution of §16), the price is $60 + 20 = 80$. The plant dumps.

Dumping happens either way, because it creates $100 - 60 = 40$ of value. The assignment of rights only decides whether residents end up $80 - 60 = +20$ or 0, and the plant $+20$ or $+100$. Reverse the numbers (harm 100, gain 60) and the river stays clean under either assignment.

Cabral draws the lesson that the Coase theorem is about efficiency, not fairness: "From an efficiency point of view, it makes no difference who owns the rights to a clean river (factory or residents). This is not to say, however, that both property rights allocations are equally fair" (Cabral, p. 99). Varian's §35.2 makes the same point in consumer theory: with quasilinear preferences the efficient

amount of the externality is independent of the assignment of property rights, and only the distribution changes (Varian, p. 670).

When the theorem fails. Each hypothesis can break:

1. *Wealth effects and limited capital.* The implications "do not hold when some of the parties have very limited capital with which to make payments". M&R would apply the analysis to a contract between General Motors and Toyota, but not "uncritically" to land tenure in a developing country or to slavery in the pre-Civil War American South (p. 39).
2. *Costly bargaining.* Haggling, delay and measurement absorb part of the gains (§22).
3. *Private information.* Efficient bargaining is an assumption of the theorem, not a result: with private values, no feasible procedure may reach the efficient y (§19).
4. *Many parties.* Free riding makes unanimous agreement harder as the group grows (§21).

Rival explanations of organization (p. 39). The transaction-costs view says arrangements exist to raise the total wealth to be shared. Against the *Marxian* view (organization reflects power and class interests), M&R cite Yugoslavia, where worker-controlled firms hired managers who organized factories much like capitalist firms. Against the *Harvard school* of industrial organization (arrangements raise market power), they read resale price maintenance as a defence against discounters free riding on the service and customer education provided by other dealers. They add that only systematic evidence can decide between the views.

Three misreadings of Coase

(1) "Property rights do not matter." They matter for the distribution, and with wealth effects for the allocation too. (2) "Externalities are never a problem." The theorem assumes efficient, costless bargaining; its practical lesson is to look for the transaction costs that block it. (3) "Coase predicts a fair outcome." It predicts an efficient y and says nothing about x.

16 Dividing the gains: the Nash bargaining solution

The value maximization principle leaves the transfers open. Two families of models close them. The *axiomatic* approach asks which division satisfies reasonable properties; the *strategic* approach (next section) writes down the bargaining procedure as a game. Varian introduces both and notes that the outcome of the Nash model depends "on how risk averse the players are and what will happen if no bargain is made", leaving its full treatment out of his book (Varian, p. 581). The statement below is the standard one (Nash 1950), not taken from the course books.

Definition: Bargaining problem and Nash solution

Two players can agree on a pair of payoffs (u_1, u_2) in a feasible set U, or fail to agree and get the *threat point* (disagreement point) $d = (d_1, d_2)$. The Nash bargaining solution is the unique rule satisfying Pareto efficiency, symmetry, invariance to affine rescaling of utilities and independence of irrelevant alternatives; it selects

$$(u_1^N, u_2^N) \in \underset{(u_1,u_2)\in U,\; u_i \geq d_i}{\arg\max} (u_1 - d_1)(u_2 - d_2).$$

With no wealth effects the frontier is the 45° line $u_1 + u_2 = V$, where $V = \max_y W(y)$ is the maximal total value. Maximizing $(u_1 - d_1)(V - u_1 - d_2)$ gives

$$u_1^N = d_1 + \tfrac{1}{2}(V - d_1 - d_2), \qquad u_2^N = d_2 + \tfrac{1}{2}(V - d_1 - d_2) :$$

each party gets its threat point plus half the *surplus* $V - d_1 - d_2$. The *generalized* Nash solution maximizes $(u_1 - d_1)^\alpha (u_2 - d_2)^{1-\alpha}$ and gives party 1 the share α of the surplus: $u_1 = d_1 + \alpha(V - d_1 - d_2)$.

Threat points and bargaining weights move the transfers, never the real decision: every solution sits on the line $u_1 + u_2 = V$, so it uses a value-maximizing y. To get a larger share, improve your outside option (d_i) or your bargaining weight (α), not the size of V. Section 22 shows why that private incentive is socially wasteful.

17 Alternating offers

17.1 Finite horizon (MWG Example 9.AA.1; Stahl 1972)

Two players divide v dollars. In odd periods player 1 offers a split and player 2 accepts or rejects; in even periods the roles swap. A dollar received in period t is worth δ^{t-1} today, $\delta \in (0, 1)$. If no agreement is reached by period T, both get nothing (MWG, p. 296).

Proposition 3 (finite horizon)

The game has a unique subgame perfect Nash equilibrium (SPNE). Agreement is reached at once, and the first proposer receives

$$v_1(T) = v\left[1 - \delta + \delta^2 - \cdots + (-\delta)^{T-1}\right] = v\,\frac{1 - (-\delta)^T}{1 + \delta},$$

which is $v\,(1 + \delta^T)/(1 + \delta)$ for odd T and $v\,(1 - \delta^T)/(1 + \delta)$ for even T. Player 2 receives $v - v_1(T)$.

Proof. Backward induction. In period T the responder accepts any offer (rejecting gives zero), so the proposer takes everything: share $s_1 = 1$. In period $T - 1$ the responder is the next proposer; rejecting is worth $\delta \cdot s_1$ to him or her in period-$(T-1)$ terms, so the proposer offers exactly that and keeps $s_2 = 1 - \delta s_1$. In general $s_{k+1} = 1 - \delta s_k$ with $s_1 = 1$, whose solution is $s_k = \sum_{j=0}^{k-1} (-\delta)^j$. At $k = T$ this is the formula. Uniqueness: in each subgame the responder accepts iff the offer is at least the continuation value (accepting at indifference, as MWG and Varian assume), and the proposer's best offer is pinned down. □

Two lessons. *Who proposes last matters*: with $T = 1$ the proposer takes everything; with $T = 2$ the second mover gets δv. *The horizon matters less and less*: as $T \to \infty$, $v_1(T) \to v/(1 + \delta)$ whichever player moves last. With $\delta = 0.6$: $v_1(3) = 1 - 0.6 + 0.36 = 0.76$, and $v_1(T) \to 0.625$.

Example: Varian's three-day game (Varian, pp. 581–582)

Alice and Bob divide \$1 over at most three days; Alice offers on days 1 and 3, Bob on day 2. Alice discounts at α per day, Bob at β. On day 3 Alice takes everything. On day 2, Bob must offer Alice α (the value of \$1 tomorrow) and keeps $1 - \alpha$. On day 1, Alice must offer Bob $\beta(1 - \alpha)$ and keeps

$$1 - \beta(1 - \alpha).$$

Each party's share rises with its *own* patience: a patient Alice (α near 1) leaves Bob little to protect on day 2.

17.2 Infinite horizon: Rubinstein (MWG Example 9.AA.2)

Remove the deadline and let the players' discount factors differ, $\delta_1, \delta_2 \in (0,1)$.

Proposition 4 (Rubinstein 1982)

The infinite-horizon alternating-offers game has a unique SPNE. Player 1 offers at once

$$x_1^* = \frac{1-\delta_2}{1-\delta_1\delta_2}, \qquad x_2^* = \frac{\delta_2(1-\delta_1)}{1-\delta_1\delta_2},$$

and player 2 accepts. With $\delta_1 = \delta_2 = \delta$, the shares are $1/(1+\delta)$ and $\delta/(1+\delta)$, the limit of Proposition 3. In Varian's notation (Alice α, Bob β): Alice $(1-\beta)/(1-\alpha\beta)$, Bob $\beta(1-\alpha)/(1-\alpha\beta)$ (Varian, p. 582).

Proof sketch (Shaked and Sutton's bounding argument, used by MWG). Let M_1 and m_1 be the supremum and infimum of player 1's SPNE payoffs in any subgame where he or she proposes, and M_2, m_2 the same for player 2 when proposing. The game looks the same from every proposing node, so these bounds are stationary.

- Player 2 rejects anything below $\delta_2 m_2$ and accepts anything above $\delta_2 M_2$. So player 1 gets at least $1-\delta_2 M_2$ and at most $1-\delta_2 m_2$ (delay only shrinks the pie): $m_1 \geq 1-\delta_2 M_2$ and $M_1 \leq 1-\delta_2 m_2$.
- Symmetrically, $m_2 \geq 1-\delta_1 M_1$ and $M_2 \leq 1-\delta_1 m_1$.

Combine: $M_1 \leq 1-\delta_2(1-\delta_1 M_1)$, so $M_1 \leq (1-\delta_2)/(1-\delta_1\delta_2)$; and $m_1 \geq 1-\delta_2(1-\delta_1 m_1)$, so $m_1 \geq (1-\delta_2)/(1-\delta_1\delta_2)$. Hence $M_1 = m_1 = x_1^*$, and likewise for player 2. The strategies "always offer the other $\delta_j x_j^*$, accept any offer of at least $\delta_i x_i^*$" attain it, and immediate acceptance follows because any delay loses value both players could share. □

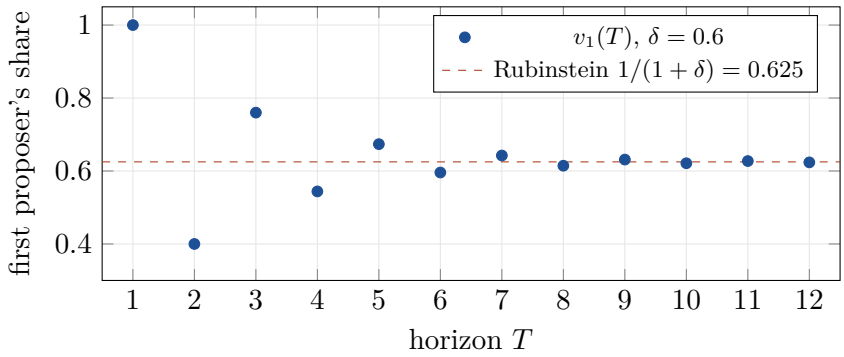


Figure 4: Finite-horizon shares oscillate (the last proposer gains) and converge to the infinite-horizon share.

What the model says about bargaining power.

1. *Agreement is immediate and efficient.* Under complete information no value is lost to delay: the threat of delay is what divides the surplus, but it is never carried out.
2. *Patience is power.* x_1^* rises in δ_1 and falls in δ_2. If $\delta_1 \to 1$ with $\delta_2 < 1$, player 1 takes everything (Varian, p. 582).
3. *The first-mover advantage vanishes as offers speed up.* Write $\delta_i = e^{-r_i \Delta}$, with Δ the time between offers. As $\Delta \to 0$, $x_1^* \to r_2/(r_1+r_2)$: the division depends only on relative

impatience. With equal rates it tends to 1/2, the symmetric Nash split, which links the strategic model to the axiomatic one of §16 (Binmore, Rubinstein and Wolinsky 1986; not in the course books).

4. *The real decision is untouched.* Discount factors move x, never y: the alternating-offers game is a model of the transfers in the Coase theorem.

Computing it. The finite-horizon recursion is three lines of Python; it reproduces the figure and, with T large, Proposition 4 (`check_part2.py` uses it):

```
def proposer_share(T, d1, d2):    # pie 1; player 1 proposes first
    share = 1.0                   # the last proposer takes everything
    who = (T - 1) % 2             # last proposer: 0 = player 1, 1 = player 2
    for _ in range(T - 1):        # one period back: the responder (who)
        d = d1 if who == 0 else d2   # must get the discounted value of
        share = 1 - d * share        # proposing next period
        who = 1 - who
    return share
```

18 Commitment, reputation and the ultimatum game

In alternating offers, power comes from what a player can credibly do next. *Commitment* changes that by removing options. M&R's examples (p. 133): in 1066 William the Conqueror burned his fleet behind his army, and Cortés repeated the stratagem in Mexico; the commitment "deterred the opposition, who then themselves retreated". When Apple built a highly specialized plant for the Macintosh, it made dropping the product more expensive, and so committed itself in the eyes of employees, competitors and customers. The cost is real: noncontractual commitment "may be worthwhile, but costly": the fleets were perfectly good, and the specialized plant may become obsolete sooner (p. 139).

Reputation as a substitute (p. 139–140). Concern for reputation checks opportunism when contracts cannot. Its value grows with the frequency of similar transactions, the horizon over which they recur, and their profitability. Hence, in relational contracts, discretion over unforeseen events should go to the party with the most to lose from a damaged reputation: the one with the longer horizon, more visibility, greater size and more frequent transactions, which in employment is usually the firm, not the employee (p. 140). Bargaining with terrorists or renegotiating executive pay advertises that there is no commitment (p. 139).

The ultimatum game (Varian, pp. 582–583). Alice proposes a division of \$1; if Bob accepts it is implemented, otherwise both get nothing. The SPNE gives Alice almost everything ("something like 99 cents"). In experiments, the average offer by U.S. undergraduates is about 45 cents and is usually accepted. Proposers are close to maximizing expected payoff given the observed rejection rates; it is the responders who depart from the theory, rejecting small offers at a cost to themselves. Explanations: social norms of fair division, which differ across cultures, and a taste for punishing unfair offers.

A one-shot offer is commitment in its purest form: the proposer is committed to walking away. A responder who could credibly commit to rejecting anything below k would receive k. A known taste for fairness or vengeance acts as exactly such a commitment, which is the question M&R pose in FFT 5.1 (pp. 163–164); FFT 5.2 (p. 164) asks what makes a signal of intent credible.

19 Private information: when efficient trade is infeasible

M&R's turn is one sentence: efficiency is defined relative to the feasible outcomes, and "what is feasible—and thus what is efficient—turns out to depend crucially on the informational conditions" (p. 140). With private information, an agreement may not be reached even when it would be efficient, and if it is reached it is not typically efficient by full-information standards, "especially when the parties have an option not to participate" (p. 140).

19.1 M&R's example: bargaining over a sale

A seller owns one unit; a buyer wants it. Each knows only his or her own value (p. 140–141):

	Seller's value \$0 (prob. 0.8)	Seller's value \$2 (prob. 0.2)
Buyer's value \$1 (prob. 0.2)	Trade	No trade
Buyer's value \$3 (prob. 0.8)	Trade	Trade

Efficient outcomes (M&R Table 5.2, p. 141). Trade is efficient with probability $1 - 0.2 \times 0.2 = 0.96$.

The temptation. A buyer worth \$3 would like to claim \$1 to pay less; a seller worth \$0 would like to claim \$2 to be paid more. Misrepresentation risks the sale falling through, but "this may be a risk worth taking" (p. 141).

The mediator device. Ask whether a neutral mediator who sets prices as a function of the two reports could make honest reporting optimal *and* trade happen exactly when efficient. If no mediator can, the parties cannot do it by themselves either (p. 141). Two kinds of constraints bind:

- *participation constraints*: nobody can be forced to trade at a loss given the value he or she reports;
- *incentive compatibility constraints*: each type must do at least as well reporting truthfully as reporting the other type.

The surplus a type must be left to report honestly is its *informational rent* (n. 7, p. 141).

Pinning down the prices (M&R Table 5.3, p. 142). A seller who claims \$2 must get at least \$2, and paying more only rewards the claim: price \$2. A buyer who claims \$1 must pay at most \$1: price \$1. One price is left, p, for reports (seller \$0, buyer \$3).

	Buyer reports \$1	Buyer reports \$3
Seller reports \$0	price \$1	price p
Seller reports \$2	no trade	price \$2

Seller worth \$0 (the buyer reports honestly):

$$\underbrace{0.2 \times 1 + 0.8\,p}_{\text{honest}} \;\geq\; \underbrace{0.2 \times 0 + 0.8 \times 2}_{\text{claim \$2}} = 1.60 \quad\Longrightarrow\quad p \geq 1.75.$$

Buyer worth \$3:

$$\underbrace{0.2 \times (3-2) + 0.8\,(3-p)}_{\text{honest}} = 2.60 - 0.8\,p \;\geq\; \underbrace{0.8 \times (3-1)}_{\text{claim \$1}} = 1.60 \quad\Longrightarrow\quad p \leq 1.25.$$

No p satisfies both. "It is impossible to find any prices that always make it in the parties' individual interests to report truthfully and to trade whenever trade is, in fact, value increasing" (p. 142).

Definition: The revelation principle (M&R n. 8, pp. 142–143)

Any pattern of outcomes achievable by any mechanism that respects incentive constraints can be achieved by a mechanism in which the parties report their values and are given incentives to report honestly. Hence restricting attention to honest reporting, as above, loses no generality: the impossibility holds for every bargaining procedure.

Reverse the probabilities (p. 143). Let the buyer be worth \$1 with probability 0.8 and the seller \$2 with probability 0.8, and set $p = 1.50$. A seller worth \$0 gets $0.8 \times 1 + 0.2 \times 1.50 = 1.10$ by reporting honestly and $0.2 \times 2 = 0.40$ by claiming \$2. A buyer worth \$3 gets $0.8 \times 1 + 0.2 \times 1.50 = 1.10$ honestly and $0.2 \times 2 = 0.40$ by claiming \$1. Both constraints hold (checking the low types and participation too, any p between 0 and 3 works), so efficient trade is attainable.

Lying about your value is cheap when trade is likely to happen anyway, and expensive when it puts a likely trade at risk. In the original example the high-value buyer and the low-value seller each face a partner who is probably willing to trade (probability 0.8), so exaggerating costs little. Reverse the odds and every misreport kills a trade 80% of the time (p. 143).

The example is a discrete instance of a general result for bilateral trade with independent private values whose supports overlap: no mechanism is efficient, incentive compatible, individually rational and budget balanced (Myerson and Satterthwaite 1983; the name is not in the course books).

20 Incentive efficiency

If incentive constraints cannot be escaped, failing the full-information benchmark is not a failure of the parties. M&R change the benchmark (p. 143):

Definition: Incentive efficiency

A mechanism is *incentive efficient* if no alternative mechanism that is feasible in the presence of the incentive constraints is preferred by all the relevant parties. By the efficiency principle, bargainers should be expected to find incentive-efficient mechanisms.

20.1 Split-the-difference bargaining

The buyer's value B and the seller's value S are independent and uniform on $[0, 1]$, each private. Both name a price at once; if the seller's ask s is below the buyer's bid b, they trade at $(s + b)/2$; otherwise both walk away (p. 143).

Small lies pay (p. 144). Let the buyer bid truthfully and a seller with $S = 0.50$ ask 0.52. Trades are lost only when $B \in (0.50, 0.52)$, probability 0.02, each losing at most \$0.01 and \$0.005 on average: expected loss $0.02 \times 0.005 = \$0.0001$. Against that, the price rises by \$0.01 whenever $B \geq 0.52$ (probability 0.48): gain \$0.0048. Net gain \$0.0047. The losses from misreporting fall on trades that "were hardly profitable anyway".

Proposition 5 (the linear equilibrium, M&R n. 9, p. 144)

There is an equilibrium in linear strategies

$$s(S) = \tfrac{1}{4} + \tfrac{2}{3}S, \qquad b(B) = \tfrac{1}{12} + \tfrac{2}{3}B,$$

and trade occurs if and only if $B > S + \tfrac{1}{4}$.

Proof. Suppose the buyer bids $b = \beta_1 + \beta_2 B$. A seller with value S who asks s trades when $B \geq (s - \beta_1)/\beta_2$ and earns

$$\pi_S(s) = \int_{(s-\beta_1)/\beta_2}^{1} \Big(\frac{s + \beta_1 + \beta_2 B}{2} - S \Big)\, dB.$$

Differentiating (Leibniz rule; at the lower limit the price equals s): $\pi_S'(s) = \frac{1}{2}(1 - \frac{s-\beta_1}{\beta_2}) - \frac{1}{\beta_2}(s - S) = 0$, so $s = \frac{2}{3}S + \frac{\beta_1+\beta_2}{3}$. Symmetrically, if the seller asks $\sigma_1 + \sigma_2 S$, a buyer with value B who bids b trades when $S \leq (b - \sigma_1)/\sigma_2$ and the first-order condition gives $b = \frac{2}{3}B + \frac{\sigma_1}{3}$. Matching coefficients: $\sigma_2 = \beta_2 = \frac{2}{3}$, $\sigma_1 = (\beta_1 + \frac{2}{3})/3$, $\beta_1 = \sigma_1/3$, so $\sigma_1 = \frac{1}{4}$ and $\beta_1 = \frac{1}{12}$. Finally $b > s \iff \frac{1}{12} + \frac{2}{3}B > \frac{1}{4} + \frac{2}{3}S \iff B > S + \frac{1}{4}$. □

How much is lost. Let $D = B - S$; on $[0, 1]$ its density is $1 - D$. Efficient trade happens with probability $\frac{1}{2}$ and creates expected gains $\int_0^1 D(1 - D)\, dD = \frac{1}{6}$. In equilibrium trade happens with probability $\int_{1/4}^1 (1 - D)\, dD = \frac{9}{32}$, and the realized gains are $\int_{1/4}^1 D(1 - D)\, dD = \frac{9}{64}$, which is $\frac{27}{32} \approx 84\%$ of the efficient total. Trade is cut almost in half, but surplus by only 16%, because the trades lost are the ones with small gains.

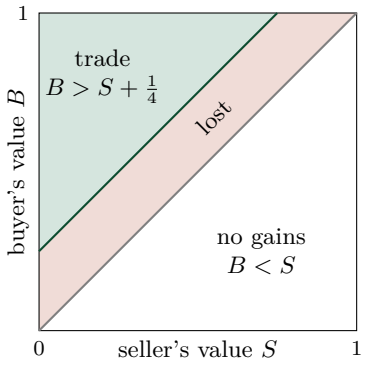


Figure 5: Split-the-difference with uniform values. Efficient trade: everything above the diagonal (probability 1/2). Equilibrium trade: the green triangle (probability 9/32). The red band $S < B \leq S + \frac{1}{4}$ holds the efficient trades lost to the incentive constraints.

M&R add that these rules "actually maximize efficiency subject to the constraints imposed by the private information" (p. 144): no other procedure does better for both. Letting the parties continue to bargain does not help; the inefficiency then shows up "either in the form of costly disagreements or as a costly delay until an agreement is reached" (p. 144). The general pattern: when incentive constraints matter, "trade takes place only if the gains from trade are sufficiently large" (p. 144).

21 Many participants: the public project

The problem worsens as more people must agree (p. 145). A project costs \$1 in total; each of N people values it at \$2 with probability p or at \$0, privately and independently. Anyone can drop

out (and lose the benefit), so nobody pays more than \$2 and someone claiming \$0 pays nothing. Value maximization says: build whenever anyone values it. If $m \geq 1$ people report \$2, build and split the cost among them.

$N = 2$. A person worth \$2 compares honest reporting with claiming \$0, given the other reports honestly (p. 145):

$$\underbrace{p\,(2-0.5)+(1-p)(2-1)}_{\text{honest}} \;\geq\; \underbrace{p\cdot 2+(1-p)\cdot 0}_{\text{free ride}} \quad\Longleftrightarrow\quad 1+0.5\,p\geq 2p \quad\Longleftrightarrow\quad p\leq \tfrac{2}{3}.$$

If $p > 2/3$ the other is too likely to pay alone, and the temptation to free ride wins (p. 146). The remedy is to build less often when the reported gains look small: build with probability q when only one person reports \$2. The constraint becomes $1.5\,p+(1-p)\,q \geq 2pq$, i.e. $q \leq \dfrac{3p}{2(3p-1)}$, which falls to $\frac{3}{4}$ as $p \to 1$ (p. 146): a quarter of the time a volunteer appears, the project is not built.

$N = 3$. With honest others, the constraint

$$p^2\left(2-\tfrac{1}{3}\right)+2p(1-p)\left(2-\tfrac{1}{2}\right)+(1-p)^2(2-1) \;\geq\; 2\left[1-(1-p)^2\right]$$

holds only for $p \leq 0.4417$ (M&R: "less than about 0.44", p. 146). At $p = \frac{1}{2}$, building with probability q when exactly one person reports \$2 requires $\frac{1}{4}\cdot\frac{5}{3}+\frac{1}{2}\cdot\frac{3}{2}+\frac{1}{4}q \geq \frac{1}{4}\cdot 2+\frac{1}{2}\cdot 2q$, i.e. $q \leq \frac{8}{9}$: the project goes ahead "less than 90 percent of the time" when one pivotal person is willing to pay for all of it (p. 146).

Large N. Lying is punished only when you are *pivotal*: when your report decides whether the project is built. As N grows it becomes nearly certain that someone else values the project, so the cost of free riding vanishes and misrepresentation cannot be prevented (p. 146). Allowing for everyone's strategic reporting, M&R report that the probability that someone volunteers tends to "approximately two chances in seven, even though it is essentially certain that funding the project is value maximizing" (pp. 146–147).

> *More participants make agreement harder, not easier. The free-rider problem is a private information problem: if values were known, the group could assign costs and build whenever efficient.*

22 Bargaining costs and investments in bargaining position

Bargaining costs are "the transaction costs of reaching mutually acceptable agreements" (p. 147). Bounded rationality imposes them even without strategic behaviour: it takes time to list contingencies, find efficient plans and settle divisions, and the resulting incomplete contracts create further costs of achieving commitment. Private information adds breakdown and delay. Hence "simple value maximization computed without regard to the constraints imposed by incentives is a misleading basis for prediction" (p. 147).

Measurement costs (p. 147–148). When the quality of a good is unknown to both sides (oil under a tract of land, the durability of equipment), each side can gain a bargaining advantage by measuring it. If one side measures, the other must too, or face adverse selection. Yet the measurement changes only the price, not who ends up with the good: if each side spends k, the good still changes hands and total wealth falls by $2k$. "The costs of gathering that information are subtracted from the parties' total wealth; there is no offsetting benefit unless the information affects a productive decision or the allocation of the asset" (p. 147).

Example: De Beers (M&R p. 148)

De Beers, which controlled over 80% of world diamond supply, sold rough stones in "sights": packets sorted on gross characteristics only, offered at a formula price on a strict take-it-or-leave-it basis, with no negotiation over price or contents, and with exclusion from future sales for any buyer who refused a sight. A monopoly reading explains part of this. The value-maximizing reading explains the details. If bargaining were allowed, buyers would examine each stone closely to argue over price, and De Beers would have to do the same in self-defence; only the final buyer, who cuts the stone, needs that examination. Forbidding negotiation removes the incentive to measure; exclusion stops "leave it" from being used as a bargaining threat; and buyers know the stones have not been picked over. Both sides save. The same logic is offered for block-booking of films, prepackaged produce, product warranties and percentage royalties for authors (p. 148).

Investing in bargaining advantages (p. 149). Measurement costs belong to a larger class: expenditures with private value because they shift the division, but no social value. M&R's example is weapons systems and the Strategic Defense Initiative funded as "bargaining chips" to be traded away in arms-control talks. Haggling, posturing and delay are wastes of the same kind, and if the parties foresee them they may never attempt the agreement at all.

Example: Preview of Topic 6: a specific investment and an ex-post split

A supplier can make a relationship-specific investment $e \geq 0$ at private cost $e^2/2$ that raises the joint value by $a\,e$. Contracts cannot specify e, and the value is split 50/50 by bargaining after the investment (the symmetric Nash solution of §16, with the investment sunk). The supplier maximizes $\frac{1}{2}ae - \frac{1}{2}e^2$ and picks $e = a/2$, while the efficient level maximizes $ae - \frac{1}{2}e^2$ and is $e = a$. Total surplus is $\frac{3}{8}a^2$ instead of $\frac{1}{2}a^2$: a quarter is lost. Ex-post bargaining is efficient given e (Coase) but distorts the investment that came before it. This is the hold-up problem (M&R Ch. 5, investments and specific assets, pp. 134–139; Varian "Hold Up", p. 579), which returns in Aula 9.

23 How the pieces fit

Hypothesis	What happens when it fails	Where
No wealth effects	the real choice depends on who holds the rights (WTP $\neq$ WTA)	§13, (p. 39)
Unlimited transfers	value-creating deals blocked by limited capital	(p. 39)
Complete information	efficient trade may be infeasible; incentive-efficient rules drop small-gain trades	§19–§20
Few parties	free riding grows with N; volunteering $\to$ about 2/7	§21
Costless bargaining	measurement, haggling, bargaining chips, delay	§22
Contractible actions	ex-post bargaining distorts investment (hold-up)	§22

The Coase theorem is a benchmark: under its hypotheses organization would not matter, because any arrangement would be renegotiated to the efficient one. Each failure in the table is a transaction cost, and the rest of the course is about the organizations designed to reduce them: the coordination and motivation problems of Part I, seen from the contract side.

24 Common mistakes

Mistake	Correction
"Bargaining power decides what gets done"	Under no wealth effects it decides only the transfers x; efficiency decides y (pp. 36–38).
"Coase: property rights do not matter"	They decide the distribution, and with wealth effects the allocation too (WTP/WTA example).
"Value maximization = efficiency, always"	Only without wealth effects; otherwise sum-maximizing is sufficient, not necessary (n. 10, p. 37).
"Rational parties always reach efficient deals"	With private information the efficient outcome can violate incentive constraints (pp. 141–142).
"A mechanism that misses some gains is inefficient"	Judge it against feasible mechanisms: incentive efficiency (p. 143).
"The revelation principle says people tell the truth"	It says honest-reporting mechanisms lose no generality (n. 8).
"The first mover always wins"	In Rubinstein the advantage vanishes as offers speed up; patience decides (§17).
"More participants make agreement easier"	Pivotality, and with it the cost of free riding, vanishes as N grows (pp. 145–147).
"Information gathering is productive"	Not when it only moves the price: measurement costs are waste (p. 147).
"De Beers' rules are just monopoly power"	They also cut wasteful measurement on both sides (p. 148).

25 Exercises (number and page only)

From `Listas/pratica-aula-01-02.md`; statements are in the books. Suggested order: top to bottom.

Exercise	Book	Page	Practises
32 RQ 3	Varian	628	contract curve vs. the outcome of bargaining
35 RQ 1–2	Varian	687	property rights, Coase with quasilinear preferences
5.1 Front yards	Cabral	117	a two-sided externality and bargaining
Q 2.1–2.2	M&R	53	value maximization; cost sharing and side payments
Q 2.3–2.4	M&R	53	proof of the principle; no wealth effects $\Rightarrow$ quasilinear
7.6 Hernan Cortez	Cabral	180	commitment
7.14 Centipede	Cabral	182	backward induction
FFT 5.1–5.2	M&R	163–164	fairness preferences; credible commitment
9.B.13, 9.B.12	MWG	302–303	offer costs instead of discounting (infinite horizon of 9.B.7); Rubinstein with $\delta_1 \neq \delta_2$
Q 5.4–5.5	M&R	165	bargaining with private values, honest-reporting prices
FFT 2.5	M&R	52	asset specificity and bargaining position
13.10	Cabral	346	ex-post bargaining and investment (hold-up)

Sources for Part II

- Milgrom, P. & Roberts, J. (1992). *Economics, Organization, and Management*. Prentice-Hall. Ch. 2 (pp. 35–39), Ch. 5 (pp. 133–149). Read from the converted text on 2026-10-13.
- Varian, H. (2014). *Intermediate Microeconomics with Calculus*, 1st ed. Norton. §30.7 and "The Ultimatum Game" (pp. 581–583), §35.2 (p. 670). The syllabus cites the 8th edition of the non-calculus book; pages may differ.
- Mas-Colell, A., Whinston, M. & Green, J. (1995). *Microeconomic Theory*. OUP. Ch. 9, Appendix A (pp. 296–299).
- Cabral, L. (2017). *Introduction to Industrial Organization*, 2nd ed. MIT Press. §5.1 (pp. 96–99).
- Outside the course books, cited for context only: Nash (1950), *Econometrica* 18(2), 155–162; Rubinstein (1982), *Econometrica* 50(1), 97–109; Binmore, Rubinstein & Wolinsky (1986), *RAND Journal of Economics* 17(2), 176–188; Myerson & Satterthwaite (1983), *Journal of Economic Theory* 29(2), 265–281.
- Every number in this Part is checked by `check_part2.py` (sympy and numpy).